



Does Syndication With Local Venture Capitalists Moderate the Effects of Geographical and Institutional Distance?



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ABSTRACT

Venture capitalists (VCs) face additional risks and costs when they invest in firms located in geographically remote countries or in countries whose institutions differ substantially from those in their home countries. Our study considers foreign VCs' prospect of overcoming these investment obstacles as a rationale for syndicating with local VCs from the investment countries. Through such syndication, foreign VCs may obtain easier access to investment opportunities, improve the risk allocation and face lower information costs. Using a novel dataset of worldwide deals, we draw a diametrically opposed picture for the two kinds of distance: our results lend support to the conjecture that the obstacles of great institutional distance cannot be overcome with the help of a local VC, whereas those of great geographical distance can.

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1. Introduction

Venture capital investment is no longer a local business in which US venture capitalists (VCs) invest in US-based entrepreneurial firms. Nowadays VCs based in the US finance a large number of entrepreneurial firms in Asia (especially China and India), Europe and elsewhere. In the last two decades, venture capital industries have developed in many countries and several VCs from these countries have also gone international. In the period 2000–2008, which our sample covers, we found that every third VC invested abroad and more than one third of worldwide venture capital investments were cross-border deals. In many of these deals VCs invested in geographically remote countries or in countries whose institutions differed substantially from those in the VCs' home countries.

The VCs' business model is based on actively monitoring and supporting typically young, risky and intransparent entrepreneurial firms (e.g. Sahlman, 1990). This requires information collection and frequent (to a large extent personal) interaction, which is likely to be more difficult and costly if the geographical and institutional distance between the VC and entrepreneurial firm (EF) is great (e.g. Wright et al., 2005), making distant investments less likely than proximate ones. Investing on great geographical and institutional distances incurs not only larger information and transaction costs, but VCs also face larger risks due to increased unfamiliarity. VCs

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may have developed certain strategies for overcoming the obstacles and reducing the risks of great geographical and institutional distance between them and the EFs they want to invest in. This paper focuses on one of these potential strategies, namely syndication with local VCs. Analyzing our dataset of worldwide venture capital deals, we find that foreign VCs syndicate with a local VC in 57% of all cross-border deals. Syndication may be chosen as a means of reducing the costs and improving the risk allocation that would burden the foreign VC if it invested alone. This is because through syndication the foreign VC may obtain easier access to investment opportunities and face lower information costs and risks.

To test whether the involvement of local VCs moderates the negative effects of geographical and institutional distance on the likelihood of investment and whether this moderating effect works differently for the geographical and institutional distance, we develop a model on the likelihood that a particular VC invests in an EF located outside its home country. To capture the various facets of institutional distance in this model, we follow institutional theory and distinguish between the regulative, normative and cognitive pillars of the institutional environment (Scott, 1995/2001), but we also use a composite measure that includes all these three pillars. We expect that the likelihood of a foreign VC's investment depends negatively on its geographical and institutional distance from the EF, as the risks and costs increase, and we hypothesize that the negative effect of geographical as well as institutional distance will be less pronounced if the foreign VC syndicates with a local VC than if it invests alone. As we will show further down, the evidence that we draw from our dataset of worldwide venture capital deals supports only the first hypothesis: syndication with a local VC makes it easier for the foreign VCs to invest in geographically distant countries but it does not moderate the negative effect of great institutional distance on the likelihood of investment. Obviously, a great institutional distance between VCs deters their cooperation to a larger extent than a great geographical distance does. While syndication with geographically distant (but institutionally close) local VCs might reduce information costs and risks for the foreign VCs, such reduction tends not to occur when foreign VCs syndicate with institutionally distant local VCs.

To the best of our knowledge, our study is the first to consider the prospect of a foreign VC overcoming the obstacles and reducing the investment risks of great geographical and institutional distance to be a rationale for syndicating with a local VC. VCs have many different reasons for wishing to syndicate their investments, most of which have been investigated in the literature mainly on the national level, while international venture capital syndication has been investigated only by a handful of studies. Among these, we should note the study by Guler and McGahan (2006), who demonstrate that US-based VCs do not syndicate more intense when they invest in non-US-based than in US-based ventures; also, that by Meuleman and Wright (2011), who focus on private equity deals from the UK to Continental Europe, rather than on cross-border venture capital deals.

Our paper adds to the research on the moderating effects of syndication and networks. An interesting study on within-country venture capital investments in the US is that by Sorenson and Stuart (2001), who consider the moderating effect of networks and suggest that the embeddedness of VCs in networks moderates the negative effects of geographical distance on the likelihood of investment within one country. In our study we consider the international dimension. This makes us possible to study simultaneously the moderating effects of syndication on institutional as well as geographical distance. Furthermore, our investigations expand past research on moderating effects in the cross-border context, which are an important topic in the literature on international entrepreneurship and new ventures (e.g. Davis et al., 2000; Ionascu et al., 2004). Much of that research concentrates on EFs that expand into foreign countries and often cooperate with local EFs (e.g. via strategic alliances or joint ventures) that help them enter new markets and thus may moderate the negative effects of geographical and institutional distance between the foreign EFs and their (new) target markets. In contrast to these studies, the moderating effects we investigate here involve a third party, namely the local VC, which we postulate to influence how geographical and institutional distances affect the likelihood of the foreign VC's investment.

This research also contributes to the emerging literature on public and corporate governance effects (Cumming and Chakrabarti, 2014) and the strand that focuses on young entrepreneurial companies going international (Zahra, 2014). Our sample companies are affected by the interplay of public and corporate governance systems. While public governance is incorporated in the countries' institutional environments, which we capture with our measures of institutional distance, corporate governance – within a given institutional environment – is pertinent to the rules within firms. Active investors, such as VCs, are supposed to make the corporate governance systems within the companies they finance more effective (Cumming and Johan, 2013). In this context, our study investigates whether the unfamiliarity of the foreign VC with the public governance system in the EF country may be alleviated by the presence of a local VCs that might contribute to improving the corporate governance mechanisms.

Finally, our study is related to the fast growing literature on international venture capital activity.² While most studies on international venture capital activity at the micro-level focus only on VCs or private equity investors from a single country (e.g. Guler and Guillén, 2010a, 2010b; Iriyama and Madhavan, 2009; Meuleman and Wright, 2011), we investigate worldwide venture capital deals by using a comprehensive deal dataset covering VCs from 48 countries. A few works in that stream of literature consider the effects of distance and show that geographical distance and certain types of institutional distance negatively affect the intensity of worldwide bilateral venture capital flows between countries (e.g. Aizenman and Kendall, 2012). The literature on international venture capital activity also points out that VCs have difficulties to implement the corporate governance mechanisms they use in

² Several studies investigate determinants behind international venture capital flows at the country level (e.g., Aizenman and Kendall, 2012; Schertler and Tykvová, 2011; Schertler and Tykvová, 2012). Other studies focus on one specific factor and its role in the internationalization process—such as networks among VCs in the US (Guler and Guillén, 2010b) and in developed countries (Tykvová and Schertler, 2011), VC's international experience (Guler and Guillén, 2010a) or the role of immigrants in US cross-border investments (Iriyama and Madhavan, 2009). A few studies deal with performance of international venture capital deals and the role played by the investor origin and syndication. Bertoni and Groh (2014) suggest that foreign investors generate additional exit opportunities for their portfolio companies. Nahata et al. (2014) and Cumming et al. (2006) relate EF country legal institutions to success. Yet other studies focus on the EF perspective; e.g. Mäkelä and Maula (2008) investigate how an EF succeeds in attracting a foreign VC.

their home country in foreign countries with different legal systems (Schoar and Lerner, 2004). We add to this literature by investigating the moderating effects of syndication.

2. Background and hypotheses

To exploit profitable investment opportunities and to diversify their portfolios geographically, VCs may invest in foreign countries either by carrying out a stand-alone deal or by joining a syndicate. In the first case, when a VC invests in a foreign EF alone, it does not have to share any profits from this investment (Wilson, 1968); however, it has to carry all associated costs and risks. VCs actively monitor and support nontransparent EFs, which face uncertain future developments. These activities enforce intensive information collection, complex contract designs and frequent interactions between the VCs and the EFs. Therefore, geographical distance and institutional differences affect the costs and risks of cross-border investment in a venture capital deal to a much larger extent than in many other, more passive types of cross-border activities, where information collection, complex contracting, frequent contact and mutual understanding are less important and risks are lower.

The extant literature supports the view that investors who are geographically close to the investment opportunity face lower risks and information costs than more distant investors (e.g. Agarwal and Hauswald, 2010; Coval and Moskowitz, 1999; Ivkovic and Weisbenner, 2005; Kang and Kim, 2008, 2010). Besides risks and information costs, travel costs, which increase with geographical distance, are also likely to play a role, because VCs interact (often in person) with their EFs. Greater geographical distance tends to make coordination more difficult (e.g. Ceci and Prencipe, 2013). Empirical research on venture capital finance suggests that it is less costly to identify (Wright et al., 2005), screen (Cumming and Johan, 2006), as well as monitor and support (Lerner, 1995; Mäkelä and Maula, 2006; Sorenson and Stuart, 2001) geographically close EFs than geographically distant ones and that geographical distance is related to investment performance (Cumming and Dai, 2010). Higher costs and higher investment risks will go hand-in-hand with higher expected returns when investors are risk-averse. Consequently, as geographical and institutional distances increase, VCs request larger returns from their investments. This reduces the number of potentially profitable investment opportunities in distant countries relatively to close countries and, thus, diminishes the likelihood of investments as distance increases. Consequently, VCs tend to invest in geographically close EFs (Cumming and Johan, 2006; Florida and Kenney, 1988; Powell et al., 2002; Sorenson and Stuart, 2001) and when they invest across borders, they prefer geographically close countries to distant ones (Aizenman and Kendall, 2012).

While the literature documents that geographical distance negatively affects VCs' investment activity, little is known about the moderating effect of local VCs on the activities of foreign VCs. Therefore, our first aim is to investigate whether the involvement of a local VC moderates the negative effect of great geographical distance on the likelihood of a foreign VC's investment in a particular deal. Mäkelä and Maula (2008) demonstrate that in cross-border syndicates the local VC usually has access to deal flow and attracts the foreign one. They also argue that local VCs typically lead the syndicate and monitor and support the EF, which is easier from a proximate than from a distant location. Thus, when foreign VCs team up with local VCs, we expect geographical distance to have a smaller effect on foreign VCs' likelihood to invest in cross-border deals than in stand-alone deals, because then foreign VCs face lower information costs and share risks and do not have to interact in person with the EF quite as often. Also, local VCs – who are familiar with investing in their home country – face lower risks. Consequently, we postulate that the need of foreign VCs to overcome the obstacles of geographical distance, which make their investments less likely, may provide a rationale for syndication with a local VC. To give an example, we expect that a UK-based VC has higher chances to invest in a deal in the US when a US-based VC also finances this EF than when the UK-based VC invests alone.

H1. Interaction between the syndication with a local VC and geographical distance

The geographical distance between a foreign VC as a potential investor and an EF affects the likelihood of this VC's investment in a cross-border deal less negatively when a local VC is involved than it does when the foreign VC invests alone.

Since VCs enter into complex contracts with EFs and are actively involved in the selection, monitoring and support of typically intransparent and risky EFs, great institutional distance, i.e. differences between public governance systems, inevitably raises information costs and investment risks and thus poses an important obstacle to cross-border activity. In order to investigate the impact of institutional distance on the likelihood of a foreign VC participating in cross-border deals we draw on institutional theory; more specifically, on Scott's distinction (1995/2001) between the regulative, normative and cognitive pillars of the institutional environment. International business scholars who have attempted to explain variations in international business strategies and operations stress the higher costs and risks of institutionally distant transactions compared to those of institutionally proximate ones (e.g. Eden and Miller, 2004). Building on Scott's distinction, Kostova (1996) defines institutional distance between two countries as the degree of difference between the regulative, normative and cognitive pillars. This three-pillar concept has been used in many empirical studies (e.g. Pogrebnyakov and Maitland, 2011).

The *regulative* pillar includes formal rules that are set, monitored, enforced and sanctioned by the state (Kostova, 1997; Xu and Shenkar, 2002). These rules shape governance structure and contract design, which set incentives for the EF's managers (Kaplan and Strömberg, 2003, 2004), and consequently play a decisive role in the EF's success (e.g. Cumming et al., 2006; Nahata et al., Forthcoming). If VCs invest in countries in which formal rules differ to a large extent from those in their home countries, they must get informed about the EF country rules and they will be limited to effectively transfer and enforce the governance structure and contract design they typically use in their home countries (e.g. Bottazzi et al., 2009). Their investment risk increases due to unfamiliarity with local rules. In other words, a great regulative distance between two countries will raise an obstacle to cross-border investments.

Because of higher costs and risks the required return will increase, decreasing the likelihood of an investment. Thus we expect that VCs are more likely to invest abroad when the regulative distance between their home country and the host country is small. Recent evidence supports this reasoning: worldwide cross-border venture capital flows are much higher between countries with similar formal rules (e.g. Aizenman and Kendall, 2012).

The *normative* pillar “is rooted in societal beliefs and norms”, which specify how things should be done (Xu and Shenkar, 2002, p. 610; see also Kostova, 1997). They define societally accepted ways to pursue valued aims. A considerable normative distance between two countries – i.e. great differences in societal beliefs, values and norms – increases unfamiliarity and impairs communication. Maintaining smooth communication between a VC and an EF is crucial to the success of an investment because VCs actively support and monitor EFs. If the EF shares similar societal beliefs, values and norms with the VC, it will be much easier for the latter to help the EF find personnel, develop strategies and technological infrastructure that both parties would accept as well as attract customers, suppliers, partners or additional financiers. Also, the management of the EF is likely to be more eager to work with the VC in this case. On the contrary, communication problems stemming from differences in societal beliefs, values and norms can undermine each party's commitment to cooperation. Consequently, a great normative distance between the countries in which the VC and the EF are based may diminish the VC's trust in the EF and impair communication, which in turn may increase costs and risks and thus reduce the VC's willingness to invest. Existing studies offer limited empirical evidence for this aspect of venture capital cross-border deals. Mäkelä and Maula (2006) show that when the normative distance is great, it is more difficult for VCs to support problematic EFs than when the normative distance is small. Several studies from other fields indicate that normative distance is important in cross-border transactions: Weber and Camerer (2003) and Weber et al. (1996) demonstrate that great normative distance is associated with higher failure rates in cross-border mergers. Further evidence suggests that normative distance also influences how banks design the loan conditions for borrowers in foreign countries (Giannetti and Yafeh, 2012) and how direct investors enter foreign markets (Kogut and Singh, 1988).

Finally, the *cognitive* pillar is associated with widely shared social knowledge and skills (Busenitz et al., 2000). The greater the amount of social knowledge and skills that the populations of two countries share, the better the communication and understanding between these two countries. Conversely, the disadvantage of greater cognitive distance is less mutual understanding and less effective communication and collaboration (e.g. Boschma, 2005). Consequently, if a VC and an EF come from countries with markedly different knowledge and skills, this will hamper understanding, communication and collaboration and diminish the chances that the EF succeeds. Thus a great cognitive distance between a VC's home country and the host country will affect negatively the likelihood of the VC investing in that country.

Our second aim is to investigate whether a joint investment with a local VC, which faces the same set of public governance system as the EF, moderates the negative effect of great institutional distance between the potential foreign VC and the EF on the investment likelihood. This will allow us to draw conclusions on whether the need to overcome the obstacles of institutional distance, which make the potential VC's likelihood to invest less likely, may provide a potential rationale for syndication with a local VC. For all facets of institutional distance we expect, for different reasons, that foreign VCs are more likely to invest in institutionally distant environments when they team up with a local VC than when they invest alone. In syndicated deals, regulative distance will hinder foreign VCs less because the local VC knows how to implement effectively both governance structures and contract design and improve the corporate governance within the firms. Hence, the information cost and risks for the foreign VC will be reduced. Foreign VCs' normative distance from the EF will be less distortive as communication between the EF and the local VC will be much easier than it would be between the EF and a foreign VC because the local VC and the EF share, at least to a certain extent, the same societal values and norms. Finally, cognitive distance will matter less since the mutual understanding between the EF and the local VC will be typically much greater than between the EF and a foreign VC because the local VC and the EF share a similar basis of knowledge and skills. Consequently we expect that, for example, a UK-based VC is more likely to invest in a deal in Spain (a country that is at great institutional distance from the UK) when a Spanish VC also finances this EF than when the UK-based VC invests alone.

H2. Interaction between the syndication with a local VC and institutional distance

The institutional (regulative, normative, cognitive) distance between a foreign VC as a potential investor and an EF affects the likelihood of this VC's investment in a cross-border deal less negatively when a local VC is involved than it does when the foreign VC invests alone.

3. Data and variables

3.1. Data on cross-border venture capital deals

We collected information on worldwide venture capital deals and all participating VCs for the period 2000–2008 from the Bureau van Dijk Zephyr database (<http://www.bvdinfo.com>). This database has international coverage and contains information on mergers and acquisitions, initial public offerings, private equity and venture capital deals. It relies on multiple sources of information, such as company press releases, stock exchange announcements, news from financial advisors, websites and so on. We chose the year 2000 as our starting point because up to that year the geographical coverage of non-European countries in Zephyr is poor. Compared to Thomson VentureXpert (TVE), the most widely used database in venture capital research, the coverage of Zephyr is slightly more limited: within the period 2000–2008 our search in the Zephyr database yielded 38,125 domestic and cross-border venture

Table 1

Cross-border and local deals.

Source: Authors' calculation from Zephyr data (2000–2008).

Panel A. Number and volume of deals carried out in each EF country						
EF country	Number			Volume (bn EUR)		
	Cross-border deals	Local deals	Cross-border to total deals (%)	Cross-border deals	Local deals	Cross-border to total deals (%)
Australia	201	92	68.6	1.231	0.444	73.5
Austria	44	45	49.4	0.396	0.125	76.0
Belgium	111	152	42.2	0.923	0.523	63.9
Brazil	12	10	54.5	1.454	1.031	58.5
Bulgaria	10	2	83.3	0.133	na	na
Canada	387	686	36.1	6.543	2.142	75.3
Chile	1	2	33.3	0.043	na	na
China	270	84	76.3	4.679	0.677	87.4
Czech Republic	7	3	70.0	0.013	na	na
Denmark	107	112	48.9	1.298	0.331	79.7
Egypt	6	1	85.7	0.234	0.030	88.6
Estonia	6	3	66.7	0.004	0.004	53.8
Finland	111	162	40.7	0.849	0.129	86.8
France	503	1140	30.6	21.483	13.553	61.3
Germany	390	620	38.6	8.499	2.693	75.9
Greece	1	9	10.0	na	0.573	na
Hong Kong	17	2	89.5	0.564	0.004	99.3
Hungary	10	6	62.5	0.093	0.005	95.1
India	184	75	71.0	5.515	0.550	90.9
Ireland	129	84	60.6	0.940	0.252	78.9
Israel	298	198	60.1	3.200	1.405	69.5
Italy	155	124	55.6	4.406	0.858	83.7
Japan	28	24	53.8	5.856	0.235	96.1
Jordan	1	1	50.0	0.106	na	na
Lithuania	7	5	58.3	0.028	0.009	75.3
Luxembourg	15	2	88.2	0.962	0.002	99.8
Malaysia	8	8	50.0	0.600	0.016	97.3
Netherlands	146	187	43.8	3.211	0.346	90.3
New Zealand	9	20	31.0	0.035	0.023	60.5
Nigeria	1	1	50.0	0.033	0.009	78.2
Norway	77	80	49.0	0.843	0.278	75.2
Philippines	2	0	100.0	0.001	0	100.0
Poland	24	13	64.9	0.043	0.010	80.7
Portugal	26	21	55.3	0.416	0.056	88.2
Russia	38	19	66.7	1.161	0.155	88.2
Saudi Arabia	0	2	0.0	0	na	na
Singapore	32	8	80.0	0.297	0.028	91.4
South Africa	9	26	25.7	0.691	0.392	63.8
Spain	117	489	19.3	5.586	4.472	55.5
Sweden	244	380	39.1	3.146	0.963	76.6
Switzerland	115	39	74.7	1.637	0.204	88.9
Thailand	5	1	83.3	0.029	na	na
Turkey	7	8	46.7	0.397	0.031	92.8
Ukraine	3	1	75.0	0.085	0.022	79.6
UK	1214	1540	44.1	14.157	6.904	67.2
US	2854	9370	23.3	53.578	117.193	31.4
Uruguay	0	1	0.0	0	0.001	0
Vietnam	5	21	19.2	0.033	0.051	39.0
Total	7947	15,879	33.5	155.432	156.728	49.8

Panel B. Number of bilateral cross-border and local VC–deal dyads for selected countries

EF country	VC country									
	Canada	China	France	Germany	Israel	Sweden	UK	US	Other	Total CB
Canada	1700	4	7	4	6	4	43	492	70	630
China	0	190	5	2	3	4	38	226	169	447
France	16	0	2876	49	6	12	232	199	271	785
Germany	1	0	49	1441	0	7	208	137	252	654
Israel	9	0	14	47	650	3	49	289	55	466
Sweden	2	0	14	9	1	709	89	45	150	310
UK	14	8	86	82	21	18	3248	663	445	1337
US	515	28	226	397	508	125	923	32,356	1543	4265
Other	17	3	159	196	8	111	492	742		
Total CB	574	43	560	786	553	284	2074	2793		

Notes: In local deals all VCs have their headquarters in the EF country. In cross-border (CB) deals at least one foreign VC invests. A dyad is a connection between each participating VC and a deal. In Panel B local dyads are depicted at the diagonal.

capital deals where VCs acquired a minority stake, while our search in the TVE database for worldwide targets involved in venture-related deals over the same period yielded 38,515 companies. Zephyr, however has two advantages compared to TVE: first, it offers more comprehensive coverage of deals carried out without the participation of a US-based VC, so it does not suffer from a “home bias” towards the US as TVE does (e.g. Aizenman and Kendall, 2012). Moreover, it includes information on the deal volume and EF industry affiliation more frequently than TVE does. These advantages have convinced us that Zephyr is much more suitable for our purposes and thus outweigh the disadvantage of slightly more limited coverage.

Having chosen Zephyr as our source, we excluded all EFs in the financial sector from the initial sample of 38,125 venture capital deals. As we only focus on collaboration among non-captive VCs, we also removed all non-venture capital investors and all governmental and corporate VCs, because governmental VCs are typically restricted to domestic investments, while the decisions of corporate VCs on which countries they should invest in are primarily driven by the parent company's existing locations and strategies. This left us with a sample of 23,826 deals. For our purposes, a *deal* can be defined as an investment by one VC (a stand-alone deal) or several VCs (a syndicated deal) in an EF. These 23,826 deals correspond to 58,377 dyads between VCs and deals. In our context, a *dyad* can be defined as a connection between a participating VC and a deal (e.g. De Clercq and Sapienza, 2010). For example, a deal financed by three VCs corresponds to three VC–deal dyads.

Panel A of Table 1 displays how the number and volume of local and cross-border venture capital deals are distributed across the EF countries in which the companies are based. In *local deals*, all VCs have their headquarters in the EF country. In *cross-border deals*, at least one VC has its headquarters outside the EF country. The US have by far the highest number of deals, with 9370 local and 2854 cross-border deals, resulting in an *internationalization share* (defined as the ratio of cross-border to total deals) of 23%. Panel B presents the number of cross-border VC–deal dyads for country pairs with the highest number of cross-border dyads; for the purposes of comparison, we display the number of local dyads at the diagonal. The figures show that the strongest connection is that between the US and the UK, with 663 dyads between US-based VCs and UK-based EFs and 923 dyads in the opposite direction.

Table 2 summarizes the internationalization and syndication patterns that emerged from our sample by displaying the numbers of cross-border deals and dyads. In total we counted 7947 cross-border deals, 4523 of which are syndicated between a foreign and a local VC, 2779 are cross-border stand-alone deals and only 645 are syndicated among foreign VCs. Thus, foreign–local syndicates account for 56.9% of cross-border deals.

3.2. Dependent variable

In order to investigate how the involvement of a local VC affects the likelihood of a foreign VC's investment in a cross-border deal, we employ a variable which is binary and equals 1 if a particular VC invests in a deal and 0 if that VC does not invest. To generate this dependent variable, which is not directly observable, we first identified existing and non-existing dyads. We then applied either a pairwise combination approach or a matching approach to specify which non-existing dyads we would consider. The pairwise combination approach, which allowed us to build *all* pairwise combinations of VCs and deals, has already been used in other studies—for example, both Corwin and Schultz (2005) and Song (2004) followed this approach in order to investigate the likelihood that a bank is invited to participate in an underwriting syndicate. The matching approach, which matches each existing VC–deal dyad to a single non-existing dyad, has been employed by Sorenson and Stuart (2001), who investigated how networks between US-based VCs affect their investment patterns.

Fig. 1 depicts existing and non-existing dyads between VCs and deals (D). It shows how we constructed this variable, which equals 1 for the foreign VCs involved in cross-border deals; these are represented by dyads VC₁–D₁, VC₂–D₂ and VC₃–D₃. We then proceeded to identify non-existing dyads. As an example, in the case of VC₁ we identified VC₁–D₂ and VC₁–D₃ as non-existing dyads. When we applied the *pairwise combination approach*, we considered all non-existing dyads. We also included VCs that do not invest in cross-border deals, such as VC₄, and set the dependent variable to 0 for all corresponding dyads. Because of the repeated inclusion of deals, we ended up with nearly 28 million observations. The number of observations for each deal depends on the country in which the deal takes place because we only include VCs from outside this country. For example, for a US deal, the number of non-existing dyads is much lower than for a non-US deal since many VCs are located in the US. By contrast, when we applied the *matching approach*, we considered only one of these non-existing dyads (for which we set the dependent variable to 0). For example, we matched the VC₁–D₁ dyad with either the VC₁–D₂ or the VC₁–D₃ dyad, depending on which deal was more similar to deal D₁ in terms of deal volume, EF age, round number, industry, and year (using propensity score matching). This approach delivers a number of observations that, theoretically, should be twice as high as the number of cross-border dyads in our sample. In practice, this number is much lower due to missing deal and EF characteristics. More precisely, we were able to include 2647 existing cross-border dyads in

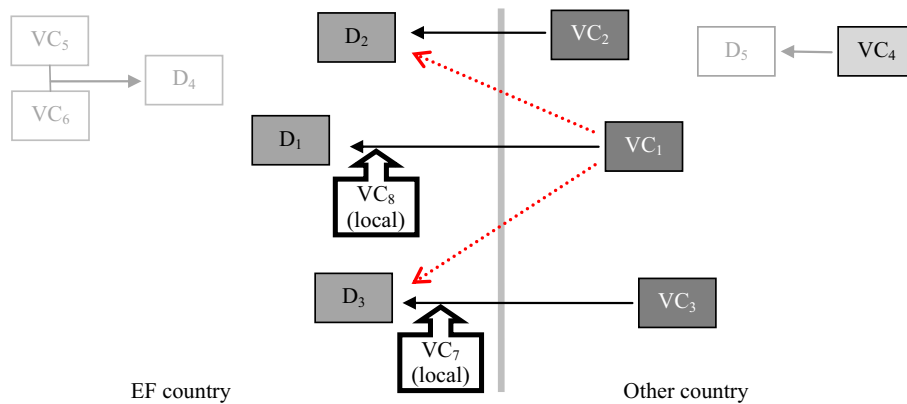
Table 2

Syndication of cross-border deals.

Source: Authors' calculation from Zephyr data (2000–2008).

	Stand-alone		Foreign–local syndicates		Foreign syndicates		Total
	No	%	No	%	No	%	No
Deals	2779	35.0%	4523	56.9%	645	8.1%	7947
Dyads	2779	11.5%	19,675	81.4%	1718	7.1%	24,172

Notes: In cross-border deals at least one foreign VC invests. A dyad is a connection between each participating VC and a deal.



Notes: This figure depicts existing dyads (solid lines) in our model of investment in cross-border deals. The various deals (D) and VCs are categorized as follows (PCA is the pairwise combination approach):



In addition, the figure depicts, for VC₁, the non-existing dyads (dotted lines), for which the binary variable of the investment in cross-border deals equals zero.

Fig. 1. Existing VC-deal dyads.

our baseline regression when we used the pairwise combination approach, but only 786 such dyads when we used the matching approach.

We use both approaches because each has its own advantages and, more significantly, theoretically outweighs a major disadvantage of the other. The major disadvantage of the *pairwise combination approach* is that the repeated inclusion of each deal can lead to a systematic suppression of estimated standard errors. Thus, with this approach, the significance of results might be overestimated. The *matching approach* does not involve the repeated inclusion of each deal, but its major disadvantage is that one might lose observations due to characteristics used in the matching routine. In our case, this disadvantage is severe, since we can consider only 30% of the dyads used in the pairwise combination approach. We compared the characteristics of dyads included in the matched sample with those not included in the matched sample and found out that the geographical and institutional distances between these two samples differ substantially. Consequently, the matching approach suffers from a sample selection problem. Therefore, we use the pairwise combination approach as our main tool of analyses.

3.3. Geographical distance and institutional distance

In order to measure the *geographical distance* between VCs and EFs we used data that identified their respective locations, i.e. zip code/postal code, city and country. Entering these data on the website BatchGeo (<http://www.batchgeo.com>), we obtained the latitude and longitude for the center point of each set of zip code/postal code, city and country. We then employed Vincenty's formula³ (1975) and calculated the distance between the centers of the two combinations of zip code/postal code, city and country for each dyad. In our main analysis we only included deals for which we were able to calculate the geographical distance between the EF and each of the participating VCs. Table 3 indicates that, on average, the foreign VC is located approximately 2700 miles from the EF irrespective of the type of cross-border deal. The median distance between a foreign VC and an EF is 1200 miles for stand-alone deals but exceeds 3200 miles when a local VC is involved which is in line with what H1 predicts. In deals in which local and foreign VCs form a syndicate, the median distance between the local VC and the EF is 22 miles. Applying the same procedure we also calculated the syndicate distance for all non-existing dyads. In our empirical analysis we used the natural logarithm of the geographical distance to capture its non-linear effect on cross-border activity (e.g. Grinblatt and Keloharju, 2001).

As a proxy for *regulative distance* we employed the absolute difference between the country-specific regulative indices of the VC country and the EF country, developed by Berkowitz et al. (2003). The regulative index is a weighted average of the following factors: efficiency of judicial system, rule of law, corruption, risk of expropriation, risk of contract repudiation, and shareholder rights and it has already been employed in the context of a cross-country study on venture capital activity by Cumming and Walz (2010). A related index (based on shareholder rights, enforcement rights and accounting standards) has been used by Nahata et al. (Forthcoming) who

³ Vincenty (1975) developed a formula for calculating geodesic distances between a pair of points on the surface of the Earth using an accurate ellipsoidal model of the Earth.

Table 3

Independent variables.

Source: Authors' calculation from Zephyr data, http://www.geert-hofstede.com/hofstede_dimensions.php, Berkowitz et al., 2003, IMD World Competitiveness Online and Orbis (2000–2008).

		Stand-alone	Foreign–local syndicates	Foreign syndicates
<i>Geographical distance between VC and EF (miles)</i>				
Distance of VCs located outside the EF country	Mean	2724	2774	2702
	Median	1205	3272	2431
	Std dev	2696	2295	2383
Distance of (all) VCs located in the EF country	Mean	–	741	–
	Median		207	
	Std dev		1024	
Distance of the (closest) local VC	Mean	–	266	–
	Median		22	
	Std dev		605	
<i>Institutional distance between VC and EF</i>				
Regulative distance ^a	Mean	1.46	1.22	1.75
	Median	0.44	0.70	0.84
	Std dev	2.06	1.48	2.04
Normative distance ^a	Mean	1.28	1.31	1.46
	Median	0.90	1.25	1.41
	Std dev	1.10	1.06	1.17
Cognitive distance ^a	Mean	1.02	0.87	1.09
	Median	0.60	0.51	0.59
	Std dev	1.06	0.90	1.17
<i>Experience and repeated relationships</i>				
Experience of VCs	Mean	82.2	24.7	25.7
	Median	17	11	7
	Std dev	114.9	46.1	50.0
Industry experience of VCs	Mean	6.1	8.0	7.8
	Median	0	3	1
	Std dev	15.7	15.4	16.5
Newcomers		48%	47%	63%
Deals with repeated relationship		0%	67%	41%
<i>EF and deal characteristics</i>				
Deal volume (million EUR)	Mean	30.8	20.7	31.9
	Median	5.2	11.7	10.0
	Std dev	132.1	78.8	160.5
Early stage		21%	44%	41%
Subsequent round ^b		38%	59%	47%

Notes: In cross-border deals at least one VC with headquarters abroad invests.

^a Of foreign VCs in cross-border deals.^b Only for EFs founded after 1999.

have found a significant effect on EF performance. The median regulative distance is 0.70 in foreign–local syndicates but only 0.44 in cross-border stand-alone deals.

To capture normative distance, we followed Busenitz et al. (2000), who state that most popular measures of the normative pillar are based on Hofstede's cultural dimensions (e.g. Ionascu et al., 2004; Jensen and Szulanski, 2004). They encompass norms, values and beliefs held by individuals in a country and include: power distance index, individualism, masculinity, uncertainty avoidance index and long-term orientation. To proxy the *normative distance* between two countries we adapted the composite four-dimensions Hofstede's distance index developed by Kogut and Singh (1988), to a five-dimension index that includes long-term orientation, a dimension that Hofstede added in 1991 in collaboration with Michael Bond. The four-dimension index has already been used in the context of cross-border venture capital flows (e.g. Mäkelä and Maula, 2006 or Nahata et al., Forthcoming). The median normative distance is 1.25 in foreign–local syndicates and 0.90 in stand-alone deals, while the mean normative distances of these two types of deals are quite similar.

Finally, with the cognitive pillar of institutional distance we capture differences in social knowledge and skills between the countries' populations. We follow Busenitz et al. (2000) who argue that the cognitive pillar is related to the prevalence of higher education. Thus, we operationalize the concept of *cognitive distance* between two countries with a three-dimension index constructed analogously to the proxy for normative distance using differences in education levels between countries (see also Ionascu et al., 2004). The three dimensions of the cognitive pillar that we consider here are secondary school enrollment, higher education achievement and total public expenditure on education. The median cognitive distance is 0.51 in foreign–local syndicates and 0.60 in stand-alone deals.

Descriptive results show that the geographical distance within some country pairs is small, while the institutional distance in one or more of the relevant pillars is great and vice versa. For example, in our sample the median geographical distance between the UK and the US is 5358 miles, while the median geographical distance between the UK and Spain is 708 miles. We observe, however, that the regulative distance between the UK and the US is only 0.44 and the normative distance is also very low at 0.31, whereas the

Table 4

Correlation coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) Log(geographical distance)	1							
(2) Regulatory distance	0.03*	1						
(3) Normative distance	0.01*	0.18*	1					
(4) Cognitive distance	0.05*	0.43*	0.20*	1				
(5) Log(1 + experience)	0.09*	−0.04*	−0.03*	−0.02*	1			
(6) Log(1 + industry experience)	0.04*	−0.09*	−0.01*	−0.04*	0.56*	1		
(7) Newcomer ^a	−0.01*	0.07*	0.04*	0.04*	−0.26*	−0.23*	1	
(8) Repeated ^a	0.00*	−0.03*	0.00*	−0.00*	0.19*	−0.22*	0.18*	1

Notes: The table reports the correlation coefficients based on the sample of the pairwise combination approach.

^a Binary variables.

* $p < 0.10$.

respective values for the UK–Spain pair are 3.28 and 1.88. Table 4, which depicts correlation coefficients in the pairwise combined sample, confirms that geographical distance is not highly correlated with either of the types of institutional distance (3.25%, 0.59%, 5.15%). However, the correlation coefficient between regulative and cognitive distance is high (42.6%). The definitions of and sources for all independent variables can be found in the Appendix A.

3.4. Control variables

We controlled for general, industry-specific and country-specific experience of potential VCs. In line with previous studies (e.g. Meuleman and Wright, 2011; Sorenson and Stuart, 2001) we measured the general and industry-specific experience of potential VCs by counting past deals. To capture country-specific experience, we generate a binary variable *newcomer*. It equals 1 if the foreign VC has not invested in the country under consideration and 0 if it has already invested in the respective country.⁴ Following Hayward (2002), who argues that although experience has a positive effect at first, as time goes by this starts to disappear, we decided to use a three-year rolling window in order to generate all variables for experience.

We expect that a potential VC has higher chances to invest in a deal if it has invested jointly with the local VC in the past (e.g. Chemmanur and Tian, 2011; Meuleman et al., 2009). For that reason we used the dummy variable *repeated*, which equals 1 if the potential VC has invested with the local VC in the previous three years.⁵ We found that 67% of all foreign–local syndicates are formed on the basis of repeated relationships (see Table 3).

With respect to deal and EF characteristics, we employed the logarithm of the *deal volume*, as well as the dummy variable *early stage* to capture whether at the time of investment the EF was at the early or the late stage of its development and a second dummy variable, *subsequent round*, which indicates whether there has been another investment round before the investment round under consideration. We constructed the last variable only for EFs founded after 1999.⁶ Table 3 shows that, syndicated deals have a higher median volume than stand-alone deals.

We also controlled for market capitalization, expected growth, innovativeness, taxation and legal environment for venture capital intermediation in the VC's and the EF's home country, on the grounds that these country characteristics may affect significantly domestic and cross-border venture capital investments (e.g. Schertler and Tykvová, 2011). Finally, we added the size of the local venture capital industry, which may affect decisions on internationalization and syndication (e.g. Tykvová and Schertler, 2011).

4. Methods and results

4.1. Econometric specifications

Below we employ two types of econometric specification, depending on whether we use the pairwise combination approach or the matching approach to specify non-existing dyads. Because, as we noted earlier, using the matching approach would have meant losing more than two thirds of the existing dyads as a result of missing deal and EF characteristics, we relied primarily on the pairwise combination approach.

Our main approach – the pairwise combination approach – generated multiple observations for each deal and therefore allowed us to model an individual deal-specific fixed effect. We used a conditional logit specification (Andersen, 1970; Chamberlain, 1980), which provides a semi-parametric estimation of a logit model without estimating the individual deal fixed effects. This approach is suitable for our purposes because our dataset contains only limited information on deal and EF characteristics, which can affect investment and syndication decisions. Deal fixed effects control for all deal characteristics (such as deal volume or the year in which the deal

⁴ Another possibility to measure country-specific experience would be a time-varying variable that captures whether foreign VCs have local offices in the country where the EF is located (at the time of the particular investment). Unfortunately, our dataset does not provide such information.

⁵ Meuleman et al. (2009) use a five-year window to obtain information on repeated relationships. However, given the short period covered by our sample, we opted for this shorter window.

⁶ For EFs founded in 1999 or earlier, we do not know whether the investment round we observe in our dataset is the first round or a subsequent round since our dataset only covers the period 2000–2008.

was closed, but also whether a local VC is involved). As each deal relates to a particular EF, deal fixed effects are perfectly collinear with all EF characteristics (such as size, age or financial characteristics). In addition, as each deal takes place in a particular industry and country, deal fixed effects are also perfectly collinear with all industry-specific characteristics (such as the market-to-book ratios of a particular industry) and all characteristics of the EF country (such as the institutional environment, deal opportunities or the stance of the capital market). Consequently, deal fixed effects nest these EF characteristics, industry-specific characteristics and EF country-specific characteristics and thus remove cross-sectional heterogeneity between deals, EFs, industries, and EF countries. Besides deal fixed effects, we employ VC country dummies in our conditional logit specifications that control for time invariant VC country characteristics.

The second econometric specification is a simple logit specification (without deal fixed effects) that we used for the matched sample. In this specification, we used all available deal and EF characteristics (i.e. deal volume, early stage, subsequent round, industry dummies) as additional control variables. We also employed year dummies and the country-specific characteristics (of EFs and VCs) to account for the multitude of effects that may affect the likelihood of a VC's investment.

4.2. Investment in cross-border deals

To test whether the local VC's involvement moderates the negative effect of distance on the likelihood of the foreign VC's investment, we interacted the geographical distance between the potential foreign VC and the EF with an indicator for the local VC's presence. Since one cannot infer from the estimated interaction term coefficients whether the local VC's presence increases or decreases the likelihood of a foreign VC's investment (Ai and Norton, 2003), we evaluate the model for different situations. In the upper part of Table 5 we show marginal effects evaluated when all variables are at their sample means, except for dummy variables, for which we switch the variable from 0 to 1. In the lower part we depict the marginal effects on the four distance variables evaluated separately for the situation where the foreign VC invests alone, and for that where the foreign VC syndicates with a local VC (all other variables remain at their means). We finally test whether geographical and institutional distance affects the likelihood of the foreign VC's investment differently when it invests alone and when it syndicates with a local VC.

We show the baseline regression results from a conditional logit specification in Column 1. To account for potential multicollinearity problems, we also ran regressions in which we considered only one of institutional distance types at a time (Columns 2–4) or included a composite measure of institutional distance (Column 5). Column 6 shows the results from a simple logit specification.

The results on geographical distance are as follows: the geographical distance between the potential foreign VC and the EF is significantly negatively related to the likelihood of this VC's investment. Given that all variables are at their means, the baseline regression reveals that the likelihood of a VC's investment in cross-border deals decreases by 0.14 percentage points when this VC's geographical distance from the EF doubles.⁷ This effect seems to be small at first glance. However, when interpreting it, one has to take into account that the unconditional investment likelihood is very low, as this regression includes 3,811,567 potential VC–deal dyads, of which 2647 are existing VC–deal dyads.

The separate evaluations of situations where the foreign VC invests with and without a local VC lend support to H1 that the local VC's involvement moderates the negative effect of geographical distance on the likelihood of the foreign VC's investment. The baseline regression reveals that when the geographical distance between the EF and the potential foreign VC doubles, the latter's likelihood of investment decreases by 0.07 percentage points when that VC syndicates with a local VC and by 0.21 percentage points when it invests alone. This difference is statistically significant. The effect of geographical distance on the likelihood of the foreign VC's investment does not change much when we include a single type of institutional distance at a time (Columns 2–4) or the composite measure (Column 5). In the simple logit (Column 6) the negative effect of the geographical distance between the foreign VC and the EF is insignificant when the foreign VC syndicates with a local VC, but it is significantly negative when the foreign VC invests alone, albeit the difference is not significant.

We turn now to the effects of institutional distance. Regulative, normative and cognitive distance between the foreign VC that represents a potential investor and the EF are negatively related to the likelihood of this VC's investment in a cross-border deal. In the baseline regression we find that the marginal effects on regulative distance and normative distance are significant and negative, while the one of cognitive distance is insignificant. When we include each type of institutional distance separately, each has a statistically significant and negative effect on the likelihood of the foreign VC's investment. We see that the separate evaluations of situations where the foreign VC invests with and without a local VC do not lend support to H2. All differences from the six regressions between the marginal effects evaluated when the foreign VCs invest alone and when they syndicate with a local VC are insignificant on any institutional distance.⁸

Overall, the findings displayed in Table 5 suggest that syndication with a local VC helps foreign VCs overcome the obstacles of great geographical distance, but not those of (any type of) institutional distance. For example, the likelihood that a UK-based VC invests in a deal in the US (a geographically distant, but institutionally close country) is higher when a US-based VC is also involved in this deal

⁷ In the regression, we use $\log(\text{distance})$. Doubling the mean distance results in $\log(2 \times \text{meandistance}) = \log(2) + \log(\text{meandistance})$. Extracting the mean distance from this expression gives $\log(2) = 0.69$, which is the change in the transformed distance variable; i.e. $\log(2 \times \text{meandistance}) - \log(\text{meandistance})$. The marginal effect amounts to -0.002 . Thus, if the distance doubles, the probability change equals approximately: $-0.002 \times 0.69 = -0.0014$.

⁸ In three unreported regressions (with each type of institutional distance included separately, as in Columns 2–4), we interacted the indicator for the local VC's presence with institutional distance instead of interacting it with geographical distance. In all three regressions the differences between both marginal effects (evaluated when foreign VCs invest alone and when they syndicate with a local VC) are statistically insignificant.

Table 5

Investment in cross-border deals.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Marginal effects evaluated at sample means</i>						
Log(geographical distance) ^a	−0.002*** (0.0003)	−0.002*** (0.0004)	−0.002*** (0.0003)	−0.002*** (0.0004)	−0.003*** (0.0005)	−0.093* (0.052)
Regulative distance	−0.001* (0.0003)	−0.001* (0.0005)				0.015 (0.025)
Normative distance	−0.001*** (0.0004)		−0.001** (0.0006)			−0.054** (0.022)
Cognitive distance	−0.000 (0.0004)			−0.001* (0.0007)		−0.038 (0.030)
Institutional distance comp.					−0.001* (0.0006)	
Log (1 + experience)	0.001*** (0.0004)	0.002*** (0.0006)	0.002*** (0.0007)	0.002*** (0.0008)	0.002*** (0.0007)	
Log (1 + industry experience)	0.002*** (0.0006)	0.002*** (0.0009)	0.002*** (0.0008)	0.003*** (0.0009)	0.003*** (0.0010)	0.052*** (0.014)
Newcomer	−0.024** (0.0100)	−0.035** (0.0139)	−0.034** (0.0156)	−0.040** (0.0171)	−0.041** (0.0172)	−0.266*** (0.033)
Repeated	0.008*** (0.0024)	0.010*** (0.0035)	0.009*** (0.0032)	0.011*** (0.0038)	0.012*** (0.0043)	0.246*** (0.032)
Log (deal volume)						0.037** (0.017)
Early stage						0.073** (0.034)
Subsequent round						−0.053 (0.034)
mcap _{VC}						−0.009 (0.049)
R&D _{VC}						0.066* (0.034)
tax _{VC}						0.003 (0.004)
vci_index _{VC}						−0.034 (0.032)
growth _{VC}						0.030 (0.033)
VSize _{VC}						0.001** (0.000)
mcap _{EF}						0.013 (0.050)
R&D _{EF}						0.089** (0.039)
tax _{EF}						−0.003 (0.004)
vci_index _{EF}						−0.020 (0.036)
growth _{EF}						−0.002 (0.038)
VSize _{EF}						0.000 (0.000)
Additional controls	Deal fixed effects and VC country dummies				Industry and year dummies	
<i>Selected marginal effects when the foreign VC invests alone</i>						
Log(geographical distance) ^a	−0.003*** (0.0005)	−0.004*** (0.0008)	−0.003*** (0.0006)	−0.004*** (0.0007)	−0.005*** (0.0009)	−0.047* (0.024)
Regulative distance	−0.002* (0.0009)	−0.002* (0.0012)				0.015 (0.024)
Normative distance	−0.003** (0.0012)		−0.004** (0.0015)			−0.052** (0.021)
Cognitive distance	−0.001 (0.0010)			−0.003* (0.0020)		−0.036 (0.029)
Institutional distance comp.					−0.002* (0.0011)	−0.036 (0.029)
<i>Selected marginal effects when the foreign VC invests with a local VC</i>						
Log(geographical distance) ^a	−0.001*** (0.0002)	−0.001*** (0.0004)	−0.001*** (0.0003)	−0.001*** (0.0004)	−0.002*** (0.0005)	−0.102 (0.064)
Regulative distance	−0.000 (0.0002)	−0.000 (0.0003)				0.015 (0.025)
Normative distance	−0.000** (0.0002)		−0.001* (0.0003)			−0.054** (0.022)

Table 5 (continued)

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Selected marginal effects when the foreign VC invests with a local VC</i>						
Cognitive distance	–0.000 (0.0002)			–0.001* (0.0004)		–0.038 (0.030)
Institutional distance comp.					–0.001* (0.0004)	
<i>Difference t-tests (local VC not present–local VC present), p-values</i>						
Log(geographical distance)	0.085*	0.073*	0.099*	0.090*	0.084*	0.388
Regulative distance	0.187	0.187				0.649
Normative distance	0.105		0.129			0.627
Cognitive distance	0.611			0.159		0.690
Institutional distance comp.					0.178	
Pseudo R ²	0.2484	0.2456	0.2451	0.2435	0.2457	0.1171
Wald test	4913.67***	5628.47***	4587.84***	4694.76***	5349.45***	196.87***
Number of obs. (VC–deal dyads)	3,811,567	3,828,187	4,061,370	4,075,952	3,811,567	1572
Prob = 0	3,808,920	3,825,535	4,058,602	4,073,179	3,808,920	786
Prob = 1	2647	2652	2768	2773	2647	786
Number of VCs	2288	2292	2305	2309	2288	254
Number of deals	2064	2068	2161	2164	2064	571
Number of countries	25	26	32	33	25	25

Notes: The table reports marginal effects where the dependent variable is the foreign VC's investment in cross-border deals. Columns 1–5 show marginal effects from conditional logit specifications where we set the deal fixed effect to zero and we use the sample from the pairwise combination approach. Column 6 depicts marginal effects from a simple logit specification using the sample from the matching approach. Standard errors (in parentheses) are clustered at EF country and year level (Petersen, 2009) except in Column 6, where we report heteroscedasticity-consistent standard errors. The independent variables are defined in the Appendix A. In addition, an indicator variable that equals 1 if the foreign VC syndicates with a local VC (0 otherwise) enters the model in form of an interaction term with the log(geographical distance).

^a Shows the marginal effect of log(geographical distance) that considers this variable and its interaction with the indicator variable for the presence of a local VC.

* p < 0.10.

** p < 0.05.

*** p < 0.01.

than when the UK-based VC invests alone. On the contrary, syndication with a local VC in Spain (an institutionally distant, but geographically close country) does not increase the likelihood that the UK-based VC invests in a deal in Spain.

The results of the control variables are in line with those documented in the recent literature. The general, industry-specific and country-specific experience of potential foreign VCs increases the likelihood of them investing in cross-border deals.⁹ We find that the likelihood that a foreign VC will invest in a deal is significantly higher if that VC has previously invested together with one of the local VCs involved in that deal than otherwise.

Thus, what we find is an asymmetry between the usefulness of the local VC's presence with respect to geographical and institutional distances. Therefore, in unreported modifications of the baseline regression, we tackled regional differences and sample limitations in order to see whether these may cause this asymmetric finding. Looking at regional differences, we first excluded all US-based VCs and EFs in order to check whether the country with the highest within-country geographical distances (where institutional distances are fixed) drives the results. We noticed that the marginal effects of all types of institutional distance become more pronounced when the US is excluded from the sample. Next, in order to gain insights into any differences in the investment of foreign VCs in cross-border deals, we separately analyzed EFs located in Europe and the US. While all types of institutional distance play a negative role in Europe, only normative distance is significant in the case of the US, indicating that differences in societal beliefs and informal norms primarily pose obstacles to foreign VCs that want to invest in the US. We find that geographical distance effects do not differ across regions.

The sample we have used so far is subject to two limitations. The first limitation is that using the addresses of individual VCs and EFs to calculate geographical distance forced us to exclude a considerable number of deals because we only included deals for which we had complete records of the addresses of the EF and all participating VCs. However, although we do not know the exact address of many VCs and EFs, we know the country of origin for nearly all of them. Therefore, we employed bilateral country distance, which we obtained via the website CEPII (<http://www.cepii.fr/anglaisgraph/bdd/distances.htm>) as an alternative measure that allowed us to consider most of the deals from the dataset. The second limitation is that measures of the institutional dimensions are unavailable for some countries, especially for developing ones. To address these two limitations we ran two further regressions. In the first one we used the bilateral country distance instead of the distance between individual VCs and EFs and we excluded institutional distances. In the second one we added the institutional distance to the bilateral country distance (thus eliminating the first limitation only), which reduced the number of countries from 48 to 32 and the number of observations from 27.7 to 23.4 million.¹⁰ The effects of

⁹ General experience is not included as a control variable when we use the simple logit specification because we match each existing dyad of a particular VC with a non-existing dyad of the same VC. The VC's general experience is thus identical in the existing and the matched non-existing dyad because both deals occur in the same year. Industry-specific and country-specific experience measures have been included because the deal in the non-existing dyad may relate to an EF in a different industry and country from those with which the EF in the existing dyad is associated.

¹⁰ The unavailability of institutional variables shrinks the number of observations only by about 10% because both the number of cross-border deals that take place in these countries and the number of VCs located there are small.

geographical and institutional distance on the likelihood of a foreign VC's investment likelihood are quite similar to those of the baseline regression; we merely observe that the negative effect of cognitive distance turns significant at the 5% level when all types of institutional distance are included in one regression, while it was insignificant in the baseline regression.

Noteworthy, in all these regressions, the marginal effect of geographical distance is stronger when the foreign VC invests alone than when it syndicates with a local VC. This result is in line with our H1 that the local VC's presence moderates the negative effect of geographical distance on the likelihood of the foreign VC's investment in a deal, although the significance levels of the difference tests are sometimes lower than those presented in Table 5. As regards institutional distance, we do not find in any of the robustness tests that a local VC helps overcome great institutional distance.

5. Summary and discussion

The paper suggests that geographical and institutional distances inhibit investment activity, but that foreign VCs have differing prospects of overcoming these distances through syndication with a local VC. The findings of this paper strongly support the hypothesis that the prospect of overcoming the obstacles of a large geographical distance between a foreign VC and an entrepreneurial firm through collaboration with a local VC provides the foreign VC with a rationale for syndicating with local VCs. We found that the geographical distance between a foreign VC and the entrepreneurial firm is less distorting when the foreign VC syndicates with a local VC than when it invests alone. This is in line with the argument that the geographical distance affects the (potential) relationship between the foreign and the local VC less strongly than the relationship between the foreign VC and the entrepreneurial firm. Albeit we expected that increasing geographical distances made any type of cooperation more difficult, we expected the relationship between two distant VCs to be easier to handle than the one between a foreign VC and a distant entrepreneurial firm, because to a certain extent VCs follow similar business models even if they come from different countries (Kaplan et al., 2003). Also, VCs form networks with other VCs and these networks often reach beyond their countries' borders (De Prijcker et al., 2009), which makes cross-border cooperation easier. Through syndication with local VCs, foreign VCs may obtain an easier deal access, improve their risk allocation and reduce information costs.

We expected but did not find similar effects in the case of the various facets of institutional distance. The involvement of a local VC in a deal does not enable foreign VCs to overcome the obstacles of a great institutional distance. Within foreign-local syndicates institutional distance appears to be much harder to tackle than geographical distance. In light of these results, the key conclusion from our paper is that syndication with local VCs is not equally effective as a means of overcoming the obstacles posed by great geographical and by great institutional distance. This finding was confirmed in robustness tests that tackled regional differences and sample limitations.

Our results thus suggest that both geographical and institutional distances inhibit investment activity, but syndication with a local VC helps only in overcoming the geographical but not the institutional distance; VCs seem to have larger difficulties to overcome institutional distances between each other. The latter finding is in line with the results of Schoar and Lerner (2004) who suggest that VCs use different corporate governance mechanisms in countries with different legal systems and that foreign VCs may have difficulties to implement the mechanisms they use in their home country in foreign countries that incur other legal systems. The cooperation and mutual understanding between distant VCs seems to be much better on great geographical distances (thus reducing costs and risks) than on great institutional distances. Great geographical distance between partnering VCs may be overcome by just buying a flight ticket. Thus, a local VC can be very beneficial to a geographically distant foreign VC. On the contrary, great institutional distance between VCs may raise obstacles to their efficient cooperation and communication. The reason is that institutionally distant VCs probably stick to differing business models and corporate governance rules and rarely share common networks. As a consequence, the involvement of a local VC may not help the foreign VC much in reducing the negative effects of great institutional distance. Alternatively, our approach might fail to pick up the moderating effect that a local VC may have on institutional distance because the short time span our data covers. Therefore, as researchers will have access to datasets covering longer periods in the future, they should again investigate whether syndication with local partners makes it easier for institutionally distant VCs to enter new countries by considering fully the dynamics of the syndication and internationalization patterns over time.

Our analysis leaves several questions open to further research. One area worth investigating further is to explore other mechanisms that may help foreign VCs in overcoming great distances. Although in our paper we focused on syndication exclusively among non-captive VCs, future research could investigate whether other local partners, such as lawyers, financial advisors, corporate VCs and government agencies may be helpful. Besides a cooperation with local VCs and other partners, foreign VCs may also reduce costs and risks inherent in distant deals by relocating companies to other destinations (e.g. Cumming et al., 2009). Another area worth investigating is the interplay and complementarity of various dimensions of foreign and local VC experience within cross-border syndicates. In our paper we took the foreign VC's perspective and controlled for its experience across different dimensions, namely general, industry-specific and country-specific experience. It would be worthwhile to investigate in what way the experience of foreign VCs is complementary to that of local VCs and whether complementarities in cross-border deals differ from those in local deals.

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Appendix A. Data description and sources

<i>Geographical and institutional distance</i>	
Geographical distance	Geographical distance between a VC and an EF, based on individual addresses, in miles (source: Zephyr).
Regulative distance	Absolute difference between the weighted average of the following regulative factors in the VC and EF countries: efficiency of judicial system, rule of law, corruption, risk of expropriation, risk of contract repudiation, and shareholder rights (source: Berkowitz et al., 2003).
Normative distance	Five-dimension Hofstede's distance index between the VC and EF countries. The index includes: the power distance index (I_1), individualism (I_2), masculinity (I_3), uncertainty avoidance index (I_4), and long-term orientation (I_5). We calculate it as: $\sum_{d=1}^5 \frac{(I_{dVC} - I_{dCO})^2}{Var(I_d)} / 5$
Cognitive distance	Var denotes each dimension's variance across all sample countries (source: http://www.geert-hofstede.com/hofstede_dimensions.php). Three-dimension cognitive distance index between the VC and EF countries. The index includes: secondary school enrollment (I_1), higher education achievement (I_2), and total public expenditure on education (I_3). We calculate it as: $\sum_{d=1}^3 \frac{(I_{dVC} - I_{dCO})^2}{Var(I_d)} / 3$
Institutional distance comp.	Var denotes each dimension's variance across all sample countries (source: IMD World Competitiveness Online). A composite measure of institutional distance between the VC and EF countries. We calculated it as: (regulative distance + 0.01) × (normative distance + 0.01) × (cognitive distance + 0.01)
<i>VC characteristics</i>	
Experience	Number of deals a VC carried out during the previous three years (source: Zephyr).
Industry experience	Number of deals a VC carried out during the previous three years in the industry of the EF (source: Zephyr).
Newcomer	Dummy variable equal to 1 if a foreign VC has not invested during the previous three years in the country in which the deal takes place, 0 otherwise (source: Zephyr).
Repeated	Dummy variable equal to 1 if a VC has invested together with one of the local VCs involved in the deal during the previous three years, 0 otherwise (source: Zephyr).
<i>EF and deal characteristics</i>	
Deal volume	Volume of the deal in million EUR (source: Zephyr).
Early stage	Dummy variable equal to 1 if the deal is an early-stage deal, 0 otherwise. We consider all deals with EFs older than three years as late-stage deals; the remaining deals are classified as early-stage deals (source: Orbis).
Subsequent round	Dummy variable equal to 1 if the particular investment round had been preceded by another investment round, 0 otherwise. This variable is only defined for EFs founded after 1999 (source: Zephyr).
<i>VC and EF country characteristics</i>	
mcap _{EF} , mcap _{VC}	Stock market capitalization/GDP in the previous year in the EF, VC country (source: Worldbank).
R&D _{EF} , R&D _{VC}	Business R&D expenditures/GDP in the previous year in the EF, VC country (source: IMD World Competitiveness Online).
tax _{EF} , tax _{VC}	Effective personal income tax rate in % in the previous year in the EF, VC country (source: IMD World Competitiveness Online).
vci_index _{EF} , vci_index _{VC}	Venture capital legal index in the previous year in the EF, VC country. Higher value is better (source: IMD World Competitiveness Online).
growth _{EF} , growth _{VC}	Expected real GDP growth rate for the next 3–5 years in the EF, VC country (source: Datastream).
VCsize _{EF} , VCsize _{VC}	Previous year's number of VCs located in the EF, VC country (source: Zephyr).

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